

# Online Retail Sales Analytics

End-to-End Data Quality Audit, Data Cleaning, and Exploratory Revenue Analysis

541,909

RAW TRANSACTIONS

401,604

CLEANED RECORDS

£8.28M

TOTAL VALIDATED REVENUE

12+ Months

TRANSACTIONAL HORIZON

|                     |                                                                     |
|---------------------|---------------------------------------------------------------------|
| Domain:             | E-Commerce & Retail Analytics                                       |
| Technical Stack:    | Python (Pandas, Matplotlib), Jupyter Notebook, Excel                |
| Primary Artifacts:  | 4 Execution Notebooks, Cleaned Dataset (online_retail_cleaned.xlsx) |
| Documentation Date: | August 2026                                                         |

## 1. Executive Summary

This documentation details an end-to-end data analytics workflow conducted on an e-commerce transactional dataset comprising **541,909 raw records**. The primary objective was to perform data understanding, rigorous data cleaning, and exploratory data analysis (EDA) to uncover sales trends, monthly volume patterns, and top revenue-generating products.

Through methodical data cleaning, non-retail transactions (accounting adjustments), non-positive quantity records (cancellations and anomalies), duplicate rows, and unidentifiable customer/description records were systematically handled. The resulting cleaned dataset contains **401,604 rows**, representing a total recorded revenue of **£8,278,519.424** spanning from 1 December 2010 to 9 December 2011.

### Key Analytics Highlights

- **Peak Monthly Revenue:** November 2011 achieved the highest single-month revenue at **£1,126,815.070** with **666,813 units sold**.
- **Lowest Full Month Revenue:** April 2011 recorded the lowest full-month revenue at **£425,222.671**.
- **Top Product:** *REGENCY CAKESTAND 3 TIER* led product revenue at **£132,567.70**.
- **Data Integrity Note:** December 2011 covers only 9 days (£341,539.43) and is explicitly categorized as an incomplete month.

## 2. Business Objective

The primary business objective of this project is to evaluate historical transactional data from an online retail store to evaluate revenue performance, quantify monthly volume dynamics, and identify top-performing retail items. Specific operational targets include:

- Audit raw transactional records to identify data quality anomalies, missing attributes, and accounting adjustments.
- Establish a transparent, reproducible data cleaning methodology tailored for retail sales performance tracking.
- Calculate exact revenue metrics (***Revenue = Quantity × UnitPrice***) across temporal and product dimensions.
- Provide verifiable data visualization and empirical interpretations for portfolio documentation.

## 3. Dataset Overview

The dataset captures all transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based online retail enterprise. The dataset consists of 8 core schema attributes:

| Column Name | Data Type / Format    | Description                                                                                         |
|-------------|-----------------------|-----------------------------------------------------------------------------------------------------|
| InvoiceNo   | Alphanumeric / String | A 6-digit integral number uniquely assigned to each transaction. Starts with 'C' for cancellations. |
| StockCode   | Alphanumeric / String | Product (item) code uniquely assigned to each distinct product.                                     |
| Description | Text / String         | Product (item) name/description.                                                                    |
| Quantity    | Integer               | The quantities of each product (item) per transaction.                                              |
| InvoiceDate | Datetime              | The day and time when each transaction was generated.                                               |

| Column Name | Data Type / Format   | Description                                                 |
|-------------|----------------------|-------------------------------------------------------------|
| UnitPrice   | Float / Numeric      | Product price per unit in sterling (£).                     |
| CustomerID  | Numeric / Identifier | 5-digit integral number uniquely assigned to each customer. |
| Country     | Text / String        | Name of the country where each customer resides.            |

## 4. Data Understanding & Quality Findings

Initial inspection of the raw dataset containing **541,909 rows** and 8 columns revealed significant data quality anomalies and missing value profiles.

### 4.1 Missing Value Distribution

| Attribute   | Missing Count | Percentage of Raw Data | Data Quality Assessment                                                |
|-------------|---------------|------------------------|------------------------------------------------------------------------|
| InvoiceNo   | 25,900        | 4.78%                  | Missing invoice identifier; prevents transaction grouping.             |
| StockCode   | 4,070         | 0.75%                  | Missing product code classification.                                   |
| Description | 4,223         | 0.78%                  | Missing item description; limits product-level aggregation.            |
| Quantity    | 722           | 0.13%                  | Missing unit quantity.                                                 |
| InvoiceDate | 23,260        | 4.29%                  | Missing temporal marker.                                               |
| UnitPrice   | 1,630         | 0.30%                  | Missing pricing point.                                                 |
| CustomerID  | 4,372         | 0.81%                  | Substantial missing customer attributes; hinders customer attribution. |
| Country     | 38            | 0.01%                  | Minor geographical omission.                                           |

### 4.2 Descriptive Checks & Outlier Analysis

Descriptive statistics performed on numeric columns demonstrated severe skewness and extreme outliers representing non-standard entries:

| Metric Column | Mean  | Median | Minimum Value | Maximum Value |
|---------------|-------|--------|---------------|---------------|
| Quantity      | 9.552 | 3.000  | -80,995       | 80,995        |
| UnitPrice (£) | 4.611 | 2.080  | -11,062.06    | 38,970.00     |

Important Initial Observations

- Unique Invoices:** Approximately 25,900 unique invoice numbers were observed in the initial dataset.
- Geographic Concentration:** Geographic distribution is heavily dominated by the **United Kingdom**.
- Negative Pricing:** UnitPrice contained negative values down to -£11,062.06.
- Negative Quantities:** Quantity contained extreme negative values down to -80,995 units.

## 5. Data Cleaning Methodology

To ensure high data fidelity for retail sales analysis, a 12-step cleaning protocol was executed in Notebook 02\_data\_cleaning.ipynb.

### 1–3. Treatment of Negative Unit Prices

Two records in the dataset contained negative UnitPrice values. Investigation revealed both were accounting adjustment records labeled "Adjust bad debt" and lacked a CustomerID. Because this analysis strictly focuses on retail sales transactions, these 2 accounting adjustment records were excluded.

### 4–8. Investigation and Removal of Negative / Non-Positive Quantities

A total of **10,624 negative-quantity records** were identified:

- **9,288 records** had invoice numbers starting with "C", strongly associated with order cancellations or returns.
- The remaining negative-quantity records were non-standard administrative/inventory entries, often exhibiting missing Descriptions, missing CustomerIDs, or a UnitPrice of 0.

For the sales-focused performance analysis, all records with **Quantity ≤ 0** were excluded.

### 9. Removal of Exact Duplicate Rows

A total of **5,268 exact duplicate rows** were identified across all columns and removed to prevent double-counting transaction volumes.

### 10–12. Handling Missing Attributes & Dataset Export

- **Missing CustomerID Removal:** Rows lacking a CustomerID were removed because customer-level tracking and coherent retail attribution require an identifiable customer code.
- **Missing Description Removal:** Rows lacking a Description were excluded because product-level performance analysis requires valid item metadata.
- **Index Reset:** The dataset index was reset following record filtration.

The final cleaned dataset contains **401,604 rows** and 8 original columns, exported as `online_retail_cleaned.xlsx`.

## 6. Exploratory Data Analysis (EDA)

Following data cleaning, a derived Revenue attribute was engineered in 03\_exploratory\_analysis.ipynb:

$$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$$

The total recorded revenue across the cleaned dataset (spanning 1 December 2010 to 9 December 2011) equals **£8,278,519.424**.

### 6.1 Monthly Revenue Performance

| Month         | Total Revenue (£) | Quantity Sold (Units) | Operational Notes     |
|---------------|-------------------|-----------------------|-----------------------|
| December 2010 | £552,372.860      | 295,177               | Full Month            |
| January 2011  | £473,731.900      | 268,755               | Full Month            |
| February 2011 | £435,534.070      | 262,243               | Full Month            |
| March 2011    | £578,576.210      | 343,095               | Full Month            |
| April 2011    | £425,222.671      | 277,730               | Lowest Complete Month |

| Month                | Total Revenue (£)     | Quantity Sold (Units) | Operational Notes                  |
|----------------------|-----------------------|-----------------------|------------------------------------|
| May 2011             | £647,011.670          | 367,115               | Full Month                         |
| June 2011            | £606,862.520          | 356,239               | Full Month                         |
| July 2011            | £573,112.321          | 361,359               | Full Month                         |
| August 2011          | £615,078.090          | 385,865               | Full Month                         |
| September 2011       | £929,356.232          | 536,350               | Substantial Growth Surge           |
| October 2011         | £973,306.380          | 568,898               | Substantial Growth Surge           |
| <b>November 2011</b> | <b>£1,126,815.070</b> | <b>666,813</b>        | <b>Highest Full Month (£1.13M)</b> |
| December 2011*       | £341,539.430          | 203,212               | Incomplete Month (Dec 1 – Dec 9)   |

### 6.2 Top 10 Revenue-Generating Products

| Rank | Product Description                | Total Revenue (£) | Product Classification / Note                  |
|------|------------------------------------|-------------------|------------------------------------------------|
| 1    | REGENCY CAKESTAND 3 TIER           | £132,567.70       | Highest Recorded Revenue Item                  |
| 2    | WHITE HANGING HEART T-LIGHT HOLDER | £93,767.80        | Retail Merchandise                             |
| 3    | JUMBO BAG RED RETROSPOT            | £83,056.52        | Retail Merchandise                             |
| 4    | PARTY BUNTING                      | £67,628.43        | Retail Merchandise                             |
| 5    | POSTAGE                            | £66,710.24        | Service / Fee Line Item (Non-Physical Product) |
| 6    | ASSORTED COLOUR BIRD ORNAMENT      | £56,331.91        | Retail Merchandise                             |
| 7    | RABBIT NIGHT LIGHT                 | £51,042.84        | Retail Merchandise                             |
| 8    | CHILLI LIGHTS                      | £45,915.41        | Retail Merchandise                             |
| 9    | PAPER CHAIN KIT 50'S CHRISTMAS     | £41,423.78        | Seasonal Retail Merchandise                    |
| 10   | PICNIC BASKET WICKER 60 PIECES     | £39,619.50        | Retail Merchandise                             |

## 7. Key Findings & Visualization Analysis

Notebook `04_matplotlib_visualization.ipynb` generated four specific Matplotlib chart artifacts to analyze revenue trajectories and volume dynamics.

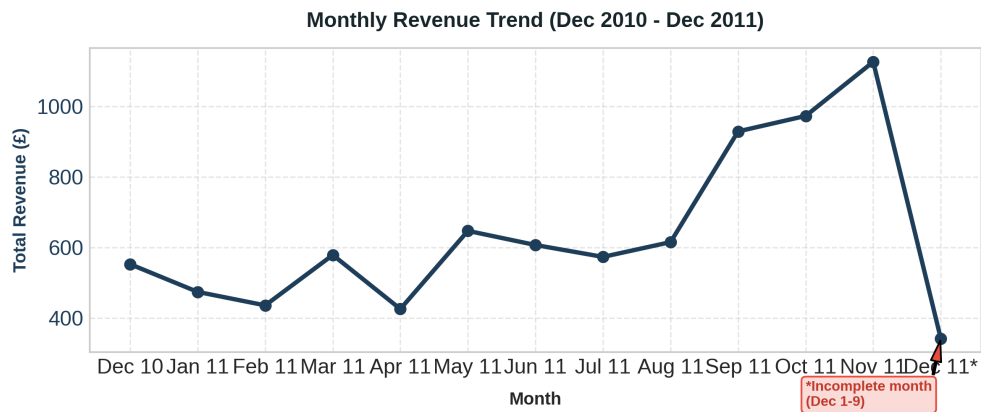


Figure 1: Monthly Revenue Trend (£) from December 2010 through December 2011.

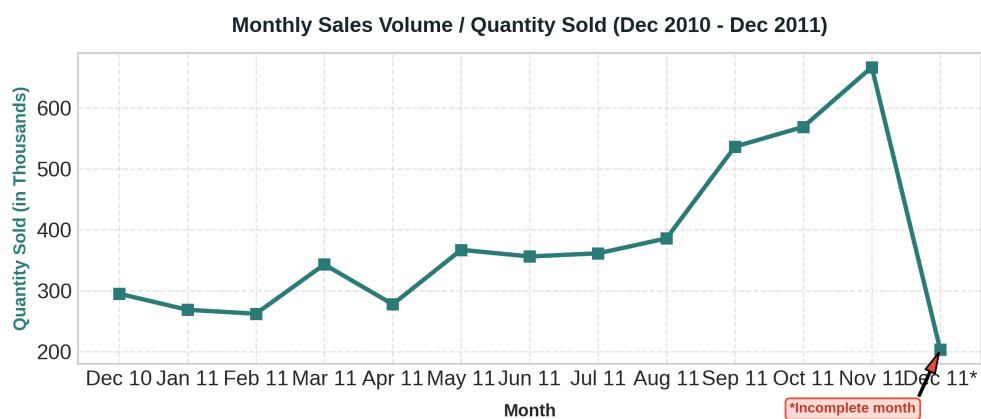


Figure 2: Total Units / Quantity Sold per Month across the clean dataset.

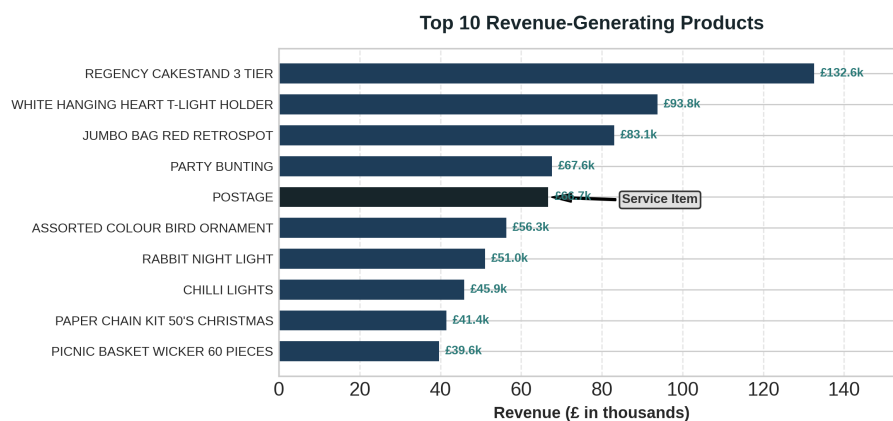


Figure 3: Top 10 Revenue-Generating Items including service item POSTAGE.

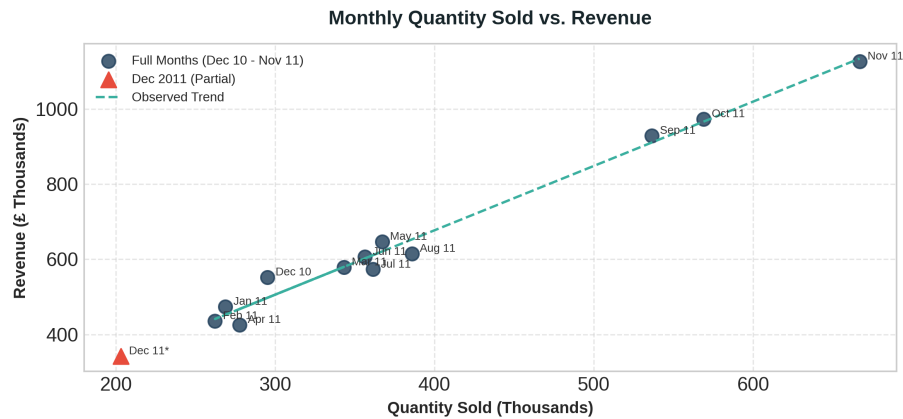


Figure 4: Scatter Plot of Monthly Units Sold vs. Recorded Monthly Revenue.

## 8. Business Interpretation & Limitations

### 8.1 Business Interpretation

- **OBSERVATION Volume-Driven Growth Trend:** Both monthly revenue and monthly sales volume exhibit a parallel, sharp upward trend from September 2011 through November 2011.
- **OBSERVATION Volume-Revenue Co-Movement:** Scatter plot visualization demonstrates that higher overall sales volume strongly aligns with higher monthly gross revenue.
- **OBSERVATION Non-Product Line Item Presence:** POSTAGE ranks as the 5th highest revenue generator (£66,710.24). In retail reporting, postage fees represent shipping charges rather than physical merchandise sales.
- **OBSERVATION Seasonal Peak:** November 2011 represents the highest operational month (£1,126,815.070 revenue; 666,813 units sold), while April 2011 represents the lowest complete calendar month (£425,222.671).

### 8.2 Analytical Limitations & Explicit Boundaries

#### Methodological & Data Constraints

- **Non-Causality:** The observed alignment between monthly unit volume and total revenue represents an empirical co-movement/relationship, **not proof of statistical causation**. External marketing spend, pricing changes, or macro factors were not measured.
- **Incomplete Data Window (December 2011):** The dataset terminates on December 9, 2011. Consequently, December 2011 (£341,539.43) covers only 9 days and **must not be directly compared** with complete calendar months.
- **Exclusion of Unidentified Transactions:** Excluding records without CustomerID or Description was necessary for customer and product attribution, but means total store-wide gross sales may be higher than reported in this customer-matched subset.
- **Exclusion of Returns:** Negative quantity records (cancellations/returns) were filtered out to evaluate positive gross sales. Net revenue (Sales minus Returns) was not evaluated in this project stage.

## 9. Project Structure & Technical Architecture

### 9.1 Sequential Execution Structure

01\_data\_understanding.ipynb

Loaded raw CSV data (541,909 rows); audited null distributions across 8 columns; conducted summary statistics on Quantity and UnitPrice.

#### 02\_data\_cleaning.ipynb

Filtered negative UnitPrice accounting entries; investigated 10,624 negative quantity records; dropped exact duplicates (5,268 rows); dropped null CustomerID and Description records; reset index; exported `online_retail_cleaned.xlsx` (401,604 rows).

#### 03\_exploratory\_analysis.ipynb

Engineered Revenue feature ( $Quantity \times UnitPrice$ ); aggregated total sales (£8.28M); computed monthly revenue and quantity tables; calculated product performance rankings.

#### 04\_matplotlib\_visualization.ipynb

Rendered 4 core portfolio charts: Monthly Revenue Trend Line, Top 10 Product Bar Chart, Monthly Quantity Line, and Scatter Plot (Quantity vs Revenue).

## 9.2 Technology Stack

| Component          | Technology / Package | Role in Project Execution                                                  |
|--------------------|----------------------|----------------------------------------------------------------------------|
| Language           | Python 3.x           | Core data manipulation and transformation environment.                     |
| Data Wrangling     | Pandas & NumPy       | Missing value auditing, Boolean indexing, aggregation, duplicate removal.  |
| Data Visualization | Matplotlib           | Custom figure generation, line plots, horizontal bar plots, scatter plots. |
| Environment        | Jupyter Notebooks    | Modular 4-step execution environment.                                      |
| Storage Format     | Excel (.xlsx)        | Cleaned dataset export target ( <code>online_retail_cleaned.xlsx</code> ). |

## 10. Conclusion & Future Scope

### 10.1 Project Conclusion

This portfolio project successfully transformed a raw, uncleaned e-commerce dataset of 541,909 records into a structured, audit-ready analytical asset of 401,604 rows. By removing accounting entries, return anomalies, duplicate rows, and incomplete identification fields, the clean dataset establishes a validated total revenue baseline of **£8,278,519.424**. The analysis successfully isolated peak sales periods (November 2011) and top-performing retail products while maintaining strict methodological boundaries between empirical observations and unsupported assumptions.

### 10.2 Future Scope

Future analytical iterations could expand upon this foundational clean dataset through the following non-overlapping extensions:

- **Return Rate & Net Revenue Analysis:** Model the 9,288 cancellation ("C") records separately to evaluate net revenue, customer return ratios, and product-level refund rates.
- **RFM Customer Segmentation:** Utilize valid CustomerID entries to perform Recency, Frequency, and Monetary (RFM) segmentation to classify retail customer cohorts.
- **Geographic Market Breakdown:** Analyze non-UK international sales performance across specific country destinations.



- **Basket Analysis:** Conduct market basket analysis (e.g., Apriori algorithm) using `InvoiceNo` grouping to identify frequently co-purchased item pairings.